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Marketing Research

Week 5

Housekeeping – Questionnaire Design
Find more details on LMS

Quiz #3 is moved to W6 (instead of W5).

Submit the following by 11 April (Friday) at 11:59 pm. To receive full points, include:

1. Research questions
2. Questionnaire in Word file (exported from Qualtrics)
   - Generate 25 survey questions and mark the corresponding scale type (N, O, I, or R). At least 15 questions are interval- or ratio-scaled.
   - Design the questionnaire using Qualtrics (https://lms.unimelb.edu.au/learning-technologies/qualtrics) and export the final questionnaire to a Word document.
   - Submit the questionnaire on time, even if your team still needs time to perfect the qualitative analysis. You will have one more chance to revise the questionnaire.
3. About your team (individual contribution)
   - Describe briefly who contributed to what up to this point. The contribution is not limited to the questionnaire design. For example, qualitative analyses.

Recap

	Label	Order	Equal Distance	Absolute 0
Nominal scale	✓			
Ordinal scale	✓	✓		
Interval scale	✓	✓	✓	
Ratio scale	✓	✓	✓	✓

Non-metric / Categorical scales

Metric / Quantitative scales

To develop a good questionnaire:

- How to phrase a question?
- How to organize the questions?
- What is the layout of the questionnaire?
- How to pre-test a questionnaire?

Topics for Today

1. Research Designs
2. Descriptive Research: Surveys
3. Causal Research: Experiments
4. Intro to Qualtrics
5. Group work time (For you to ask questions and work on your questionnaire)

Descriptive Research:

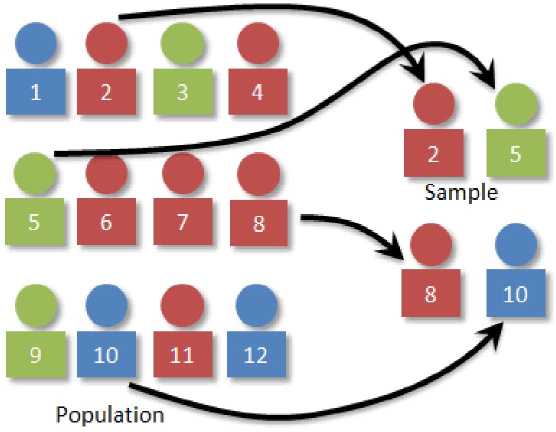
Surveys

What is a survey?

- Ask respondents for information using verbal or written questions
- Questions are fixed and structured
- Respondents are a representative sample of “target population”
- Data are mostly quantitative (e.g., multiple choice)

Useful when:

- The respondents know the answer
- We care what they have to say
- The respondents are willing to tell us the truth



Steps in Designing Survey

1. Set the survey goal

2. Determine type of questionnaire and method of administration

3. Design the items

4. Set the scale

5. Design the questionnaire

6. Pretest the questionnaire

7. Execution

Step 1. Set the Survey Goal

Determine analyses required for study	Experiments typically require t-test, Chi-Square, or ANOVA.
Determine data required to do the analyses	Consider sample size, measurement scales needed (e.g., nominal data cannot be used in many types of statistical analysis), and variables needed (e.g., for experiments, variables with many levels cause problems).
Consider the type of information or advice you want to give based on the survey	If you aim to reduce waiting time by 20%, measure waiting times in minutes to understand if goal of the study is achieved.

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Step 2. Determine Type of Questionnaire and Method of Administration

Qualtrics

	Face-to-face	Web surveys	Telephone (CATI)	Mail surveys
Costs	○	●	◑	◔
Max. Length	●	◑	○	◕
Interviewer Bias	○	●	◕	●
Sample Bias	○ - ●	◔	◕	◑
Open Answers	●	◔	●	○
Complexity (Filter)	◔	●	◕	◔
Flexibility of methods	◕	●	◔	◑

Researchers view: ○ ◔ ◑ ◕ ●

Very bad … Very good

Step 3. Design the Items

Can all the respondents answer the question asked?

- Ensure that all potential respondents can answer all items. If they cannot, ask <u>screener questions</u> to direct them. If the respondents cannot answer questions, they should be able to skip them.

Can the respondents construct/recall answers?

- If no, use other methods to obtain information (e.g., secondary data or observations).
- Ask the respondents about major aspects that occurred before zooming in on details.

Do the respondents want to fill out each question?

- If the questions concern “sensitive” subjects, check whether they can be omitted. If not, stress the confidentiality of the answers.
- social desirability bias: respondents might choose to select a position that they believe society favors.

Should you use open-ended or closed-ended questions?

- Keep the subsequent coding in mind. If easy coding is possible beforehand, design a set of exhaustive answer categories.
- Remember that open-ended scale items have a much <u>lower</u> response rate than closed-ended items.

Step 3. Design the Items (cont.)
When designing the questions, avoid

Negations: All in all, wasn't I satisfied with my holiday accommodation?

Vague quantifiers: Do you book holidays frequently?

Suggesting answers: Do you agree with the Lonely Planet Guide that this is the best hotel in Greece?

Double-barreled questions: How satisfied were you with the meals and the holiday accommodation?

Questions not answerable: Did you like the holiday accommodation booked? (subject did not book a holiday)

See W3 for more examples (e.g., leading questions, loaded questions)

Step 4. Set the Scale

Likert scale

I consider Oddjob Airlines to be....

	Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
down-to-earth	●	○	○	○	○	○	○
family oriented	○	○	●	○	○	○	○
honest	○	●	○	○	○	○	○

Imbalanced scale:

Generally, I am not satisfied with my iPhone

Completely disagree	Disagree	Neutral	Completely agree
○	○	○	○

Step 4. Set the Scale (cont.)

Set the Scale

Semantic differential scale

I consider Oddjob Airlines to be....

serious	○	○	○	○	○	◉	○	fun
expensive	○	○	○	○	◉	○	○	affordable
established	○	○	○	○	○	◉	○	new

Rank order scale

Please rank the following airlines in terms of preference

- Oddjob Airlines 1
- British Airways 2
- Easyjet 3

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Step 4. Set the Scale (cont.)

Visual Analogue scale

How high are your expectations that…

Very low — Very high

0 10 20 30 40 50 60 70 80 90 100

… with Oddjob Airways you will arrive on time. ☐ Not Applicable

… the entire journey with Oddjob Airways will occur as booked. ☐ Not Applicable

… in case something does not work out as planned, Oddjob Airways will find a good solution. ☐ Not Applicable

Step 4. Set the Scale (cont.)

Design The Questionnaire

Poor design:

	Strongly disagree	Somewhat disagree	Neutral	Somewhat agree	Completely agree
I am satisfied with the performance and reliability of my Apple iPhone	❑	❑	❑	❑	❑

Step 4. Set the Scale (cont.)

Design The Questionnaire

Poor design:

	Strongly disagree	Somewhat disagree	Neutral	Somewhat agree	Completely agree
I am satisfied with the after-sales service of my Apple iPhone	❑	❑	❑	❑	❑

Step 4. Set the Scale (cont.)

Design The Questionnaire

Poor design:

	Not at all important	Not important	Neutral	Important	Very important
Which of the following iPhone features are most important to you?					
Camera	❑	❑	❑	❑	❑
Music player (iPod)	❑	❑	❑	❑	❑
App store	❑	❑	❑	❑	❑
Web browser (Safari)	❑	❑	❑	❑	❑
Mail client	❑	❑	❑	❑	❑

Step 4. Set the Scale (cont.)

What scales to use for close-ended questions?

- Likert scales
- Semantic differential scales (label end points)
- Rank order scales

- (Usually) 5 to 7-point scales

Should you use a balanced scale?

- A **balanced scale** has an equal number of positive and negative scale categories. The wording in a balanced scale should reflect equal distances between the scale items. This is called an **equidistant scale,** which is necessary for factor analysis.

Should you want to use a forced-choice or free-choice scale?

- Forced-choice scale (e.g., 4 or 6 points without neutral category)

- free-choice scale has neutral category → lower response bias

Should you include an "undecided"/"don't know" category?

- Only for questions that the respondent might genuinely not know, should a "don't know" category be included. If included, place this at the end of the scale.
- Avoid in an attitude/preference question

Step 4. Set the Scale (cont.)

#fail!

Part 4. Following questions are related to demographic characteristics. Please answer that corresponds you.

4.1.3. What type of city do you live in?

- Big City (More than 10 million population)
- Middle to Small City (5 million to 10 million population)
- Rural area (Less than 5 million population)

Step 5. Designing Questionnaires

1.Screeners or classification questions come first. These questions determine which parts of the survey a respondent should fill out.

2.Next, **insert the key variables of the study**. This includes the dependent variables, followed by the independent variables.

3. Use the funnel approach. That is, ask questions that are more general first and then move on to details. This makes answering the questions easier as the order aids recall. Make sure that sensitive questions are put at the very end of this section.

4. **Demographics** are placed last if they are not part of the screening questions.

Step 5. Designing Questionnaires

Increasing Response Rates

It is becoming increasingly difficult to get people to fill out surveys. This may be due to over surveying, dishonest firms that disguise sales as research, and a lack of time.

Ways to overcome low response rates:

1. Send out a pre-notice letter indicating the importance of the study and announcing that a survey will be sent out shortly.
2. Send out the survey with a sponsor letter again, indicating the importance of the study.
3. Follow up after three to four weeks with both a thank you note (for those who responded) and a new survey plus a reminder (for those who did not respond).
4. Call or email those who have not responded still and send out a thank you note to those who replied in the second round.

Step 6. Pretest The Questionnaire

- You can pretest questionnaires in two ways.
- You can use a few experts (say 3–6) to read the survey, fill it out, and comment on it. Experienced market researchers can spot most issues right away and should be employed to pretest surveys.
- If you aim for a high-quality survey, you should also send out a set of preliminary (but proofread) questionnaires to a small sample of 50–100 respondents.
- Think about response rates: an incentive could be to give the participants a chance to win a product or service!
- Reporting the findings to the participants is an incentive that may help participation (particularly in professional settings).

- ***Never skip pretesting due to time issues, since you are likely to run into problems later!***

Step 6 and 7. Pretest vs Execution

Pretest…

- in the direct environment
- with the (relevant) stakeholders
- with subjects from the population

When the questionnaire is in the field…

- check the intermediate results regularly
- monitor the agency/field institute
- check the duration of the survey

Step 7. Execution

- Web surveys: use Qualtrics, LimeSurvey, Google Forms, SurveyMonkey etc.
- Take a look at how the data will look & if they can be used.
- Try the survey a few times yourself! (and delete this data so that you do not bias the results)

Descriptive Research: Type of Studies

Cross-sectional studies:

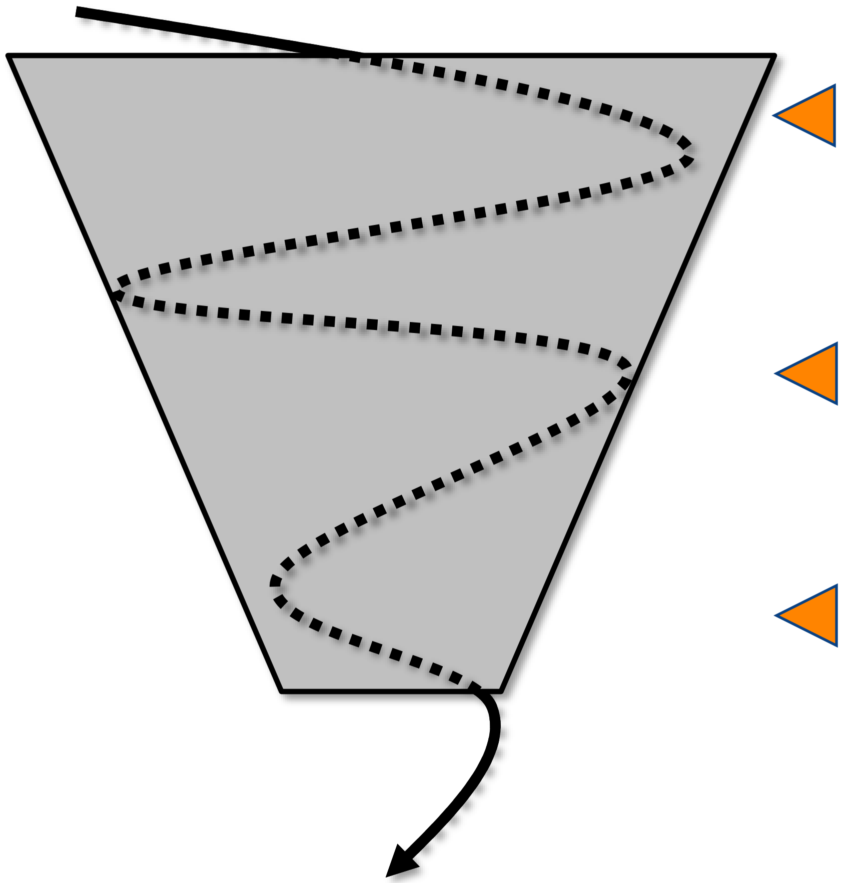
- Measure a sample of the population at only **one point in time**
- Measure the **current status** of a target market or a problem
- Most commonly used

Longitudinal studies:

- Measure the same sample of the population **multiple times over a period of time**
- Measure the **changes** in a target market or a problem
- Less common, hard to execute
- Can be supplemented by historical secondary data

Causal Research: Experiments

Determine the Research Design



Ambiguous problems:
Exploratory research design

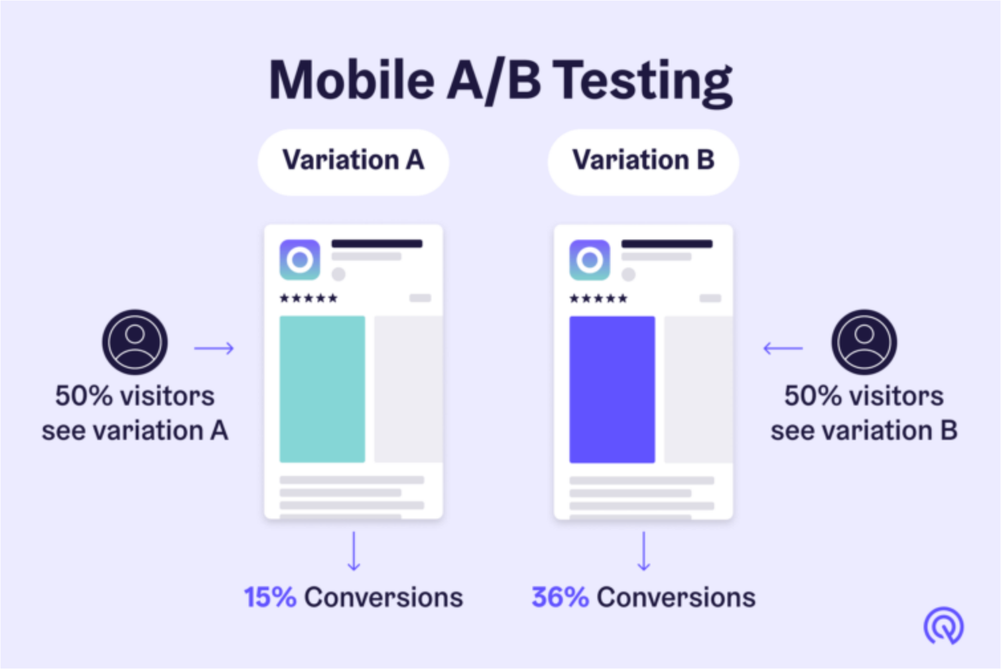
Somewhat defined problems:
Descriptive research design

Clearly defined problems:
Causal research design

Causal research

We speak of a causal effect when the effect is dependent on some value of the cause.

Let’s think…..

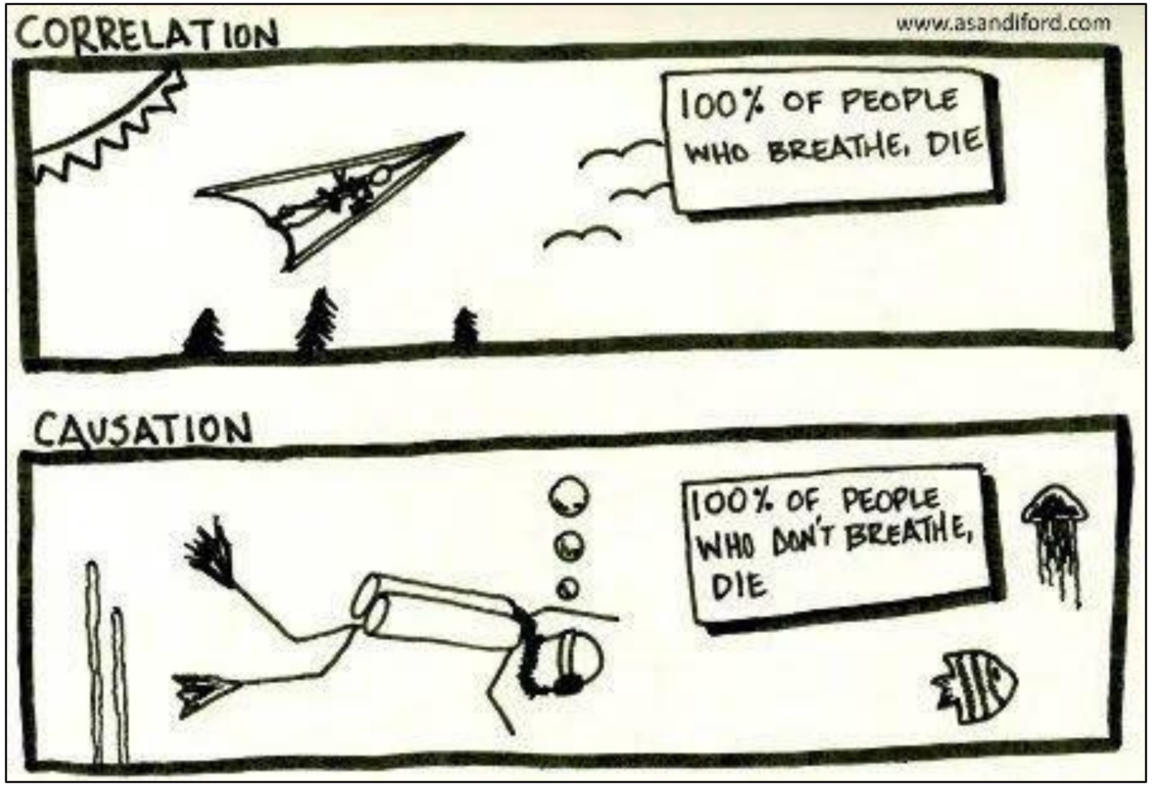


Causal research

Causal conditions:

- Relationship between cause and effect,
- time order,
- control for other factors, and
- an explanatory theory.

Is correlation same as causation?



https://www.thesatirist.com/cartoons/correlation-vs-causation.html

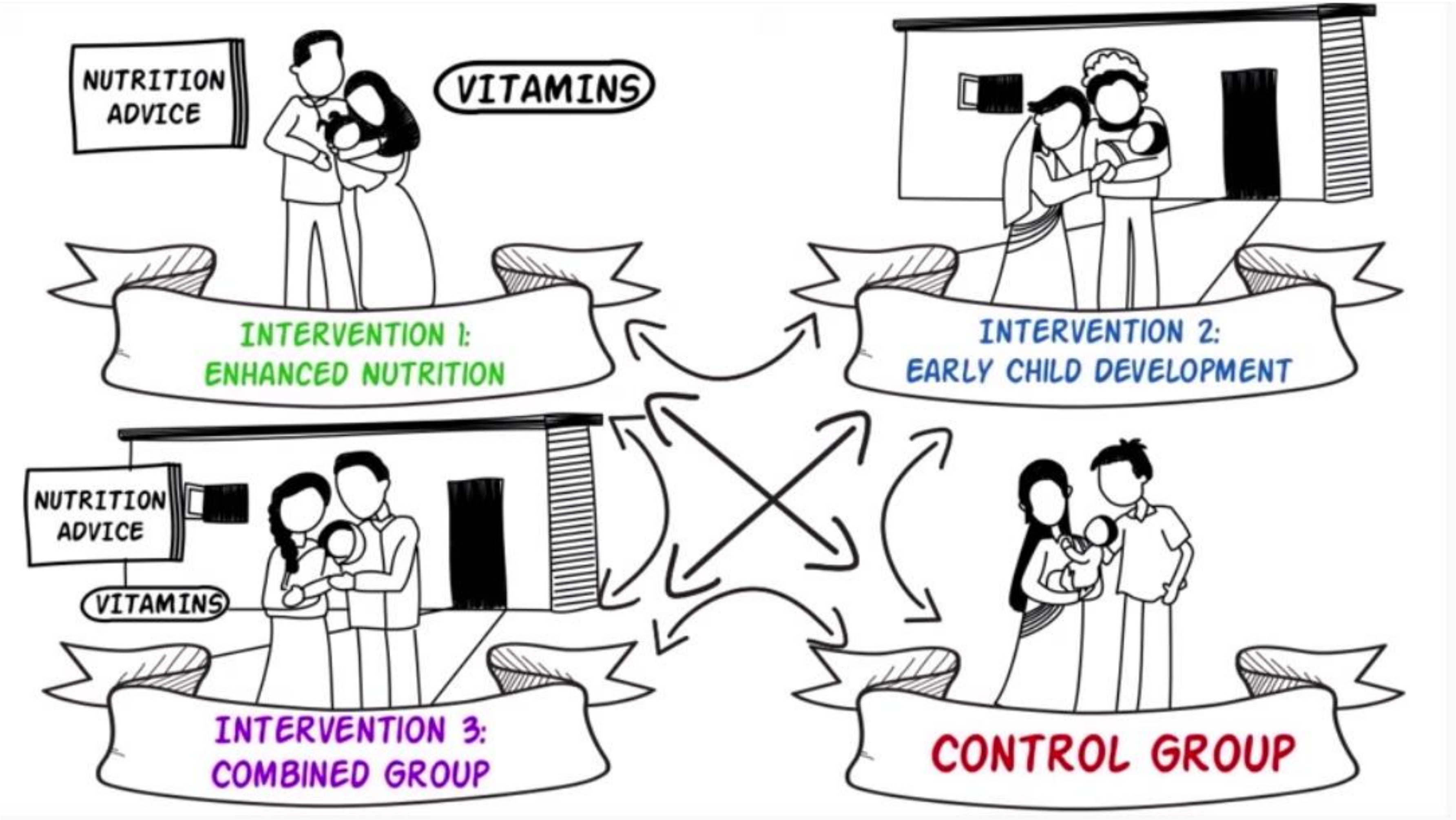
Experiments

One way of establishing causality is via experiments; study designs commonly used in causal research in which a researcher controls for a potential cause and observes corresponding changes in hypothesized effects via treatment variables.

Some forms

- Lab experiments
- Field experiments
- Test markets

Experimental Design: Example



Basic Experimental Research

- **Dependent variable(s):** what you try to explain using experiments
- **Independent variables:** what you use to explain the outcome
  - Independents are typically manipulated through *treatments* or *stimuli*.
  - The number of treatments are the *factors* or *levels* of the experiment.
- **Extraneous variables:** those variables that are part of the study but not manipulated.

Notation:

- O: A formal **observation** or measurement of the dependent variable. Subscripts such as O_1, O_2, etc. are used to indicate measurements in different points in time.
- X: Test participants' exposure to an experimental manipulation or treatment.
- R: Random assignment of participants. Randomization ensures control over extraneous variables and increases the reliability of the experiment.

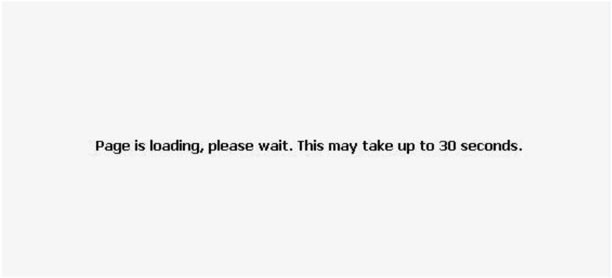
Study Designs

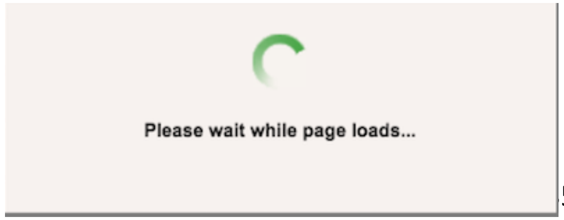
One-shot case study:		X	O_1	
Before-after design:		O_1	X	O_2
Before-after experiment with a control group:	Experimental group (R)	O_1	X	O_2
	Control group (R)	O_3		O_4

Example

- Hypothesis: Darker background colour is more likely to keep consumers on the waiting page than light background colour.
- Independent variable: Darker/light background colour
- Dependent variable: Likelihood of staying on the waiting page.

VS





One-shot case study:

One-shot case study:	X	O_1

Loading…Please Wait

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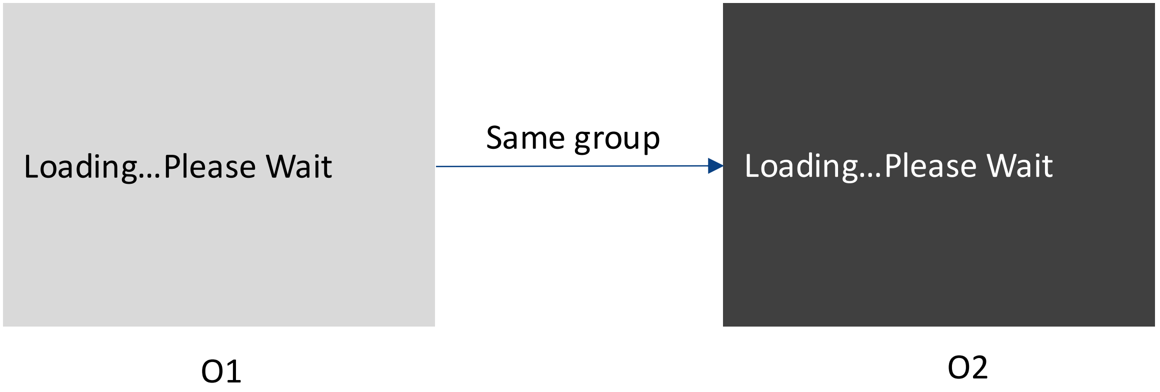
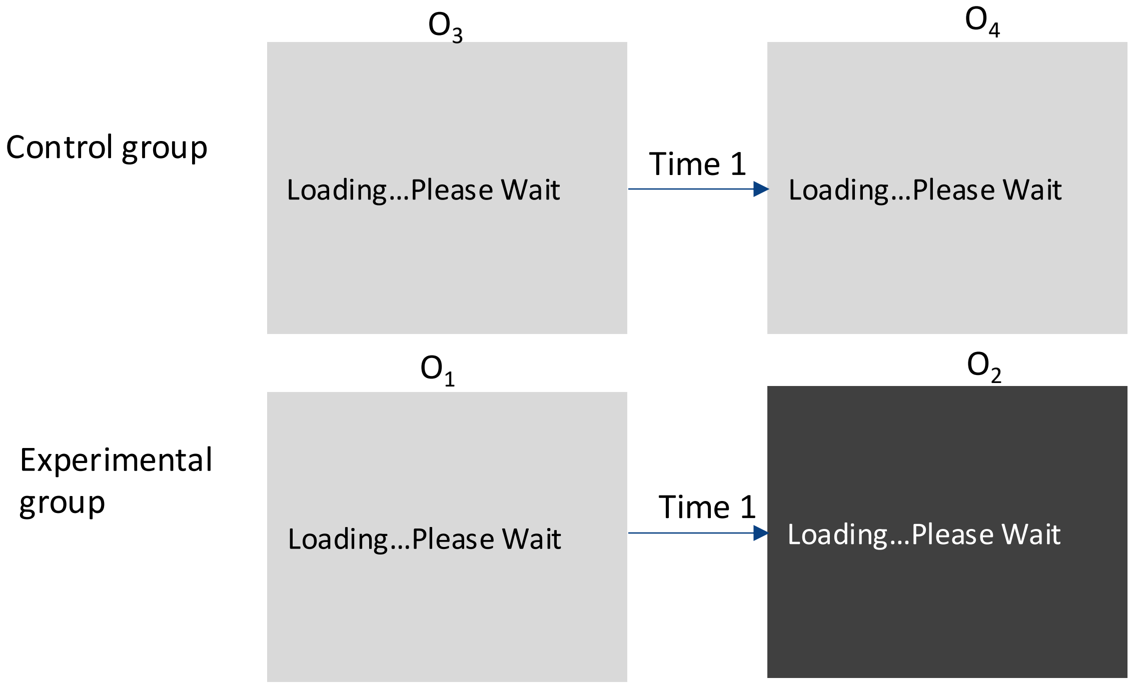
Before-after design

Before-after design: O_1 X O_2



Before-after experiment with a control group

Before-after experiment with a control group:	Experimental group (R)	O_1	X	O_2
	Control group (R)	O_3		O_4



Field experiments

Field experiments are experiments that are conducted in a field setting.

Several reasons to conduct these

- External validity
- Give some indication of expected effect

Examples: *To determine the sales potentials for new products or to test the effects of marketing mix actions*

Experiments and Validity

Internal validity: Ability to confidently draw causal conclusions

- How clean is the experiment? Can you eliminate alternative explanations (i.e., an extraneous variable)?
- Depends largely on the procedures of a study and how rigorously it is performed.
- Potential threat
  - Attrition: over the course of a longer study, participants may drop out (if caused by the treatment)

External validity: Ability to generate from the research setting to the other contexts (i.e., the real world)

- Do outcomes apply to practical situations?
- Relates to how applicable the findings are in the real world.
- Potential threat
  - Sampling bias: participants in the study differ from the population

It is difficult to achieve both high internal and high external validity at the same time.

Lab vs. Field Experiments

Lab Experiments

- Artificial setting
  - e.g., Lab's subject pool
  - limited effects from extraneous variables
- Pro:
  - Quick and inexpensive.
- Con:
  - lack of natural setting (generalization might be a problem)

Field Experiments

- Natural setting
  - e.g., supermarkets, online
- Pro:
  - Generalizable
- Con:
  - expensive and time-consuming
  - difficult to control the impact of extraneous variables in natural setting
  - no secrecy

Test Markets

Test markets are a type of field experiment that evaluates a new product or promotional campaign under real market conditions.

Marketing mix variables (independent variables) are varied in test marketing, and the sales (dependent variables) are monitored so that an appropriate marketing strategy can be identified

Common objectives:

- To determine market acceptance of the product
- To test alternative levels of marketing mix variables

- Better to invest in conducting test marketing in order to avoid the more expensive mistake of unsuccessful product rollout